The historical CH density instrument

Control: Match of Historical to Modern Geographic Units — Historical CH density index (State level).

1. Describe historical density within a modern state. Let P_{jt} be the population of historical reporting area j in census year t, and a_{jt} its mapped area in square kilometres. A reporting area is usually one historical county; counties with a joint population report form one combined area. Only source areas in the selected historical states or states-and-territories scope enter.

Population and area are split across modern states in proportion to geographic overlap, using the same allocation as area-weighted population. Let s_{jg} be the fraction of source area j inside modern state g. A source entirely inside one state has share one. Splitting leaves the historical source density P_{jt}/a_{jt} unchanged because population and area receive the same share.

2. Average log densities using population shares. Using source areas with known positive population and usable area, the instrument is

$$z_{g,t} = \frac{\sum_j s_{jg} P_{jt} \log(P_{jt}/a_{jt})}{\sum_j s_{jg} P_{jt}}.$$

A source with twice the assigned population receives twice the weight. Recorded zero-population areas receive zero weight and do not require taking the logarithm of zero. The assigned population total must be positive. Unknown populations are omitted from both sums; they are not set to zero.

3. This is separate from the modern CH index. The instrument uses historical population and geometry, with no estimated modern parameter. It enters the regression directly; the Levels–Natural log selector does not transform it again. Negative values can be valid. The modern CH model combines current county employment and land area and estimates the elasticity $\rho = \theta - 1$.

4. Coverage and identification remain necessary. A state needs at least 80% coverage by assigned areas with known population and usable mapped area. Zero-population areas count toward coverage; unknown-population areas do not. Split areas count in proportion to the allocated share. Missing or unmapped source records remain flagged but do not separately exclude a state that meets the coverage rule. The same rule applies in both selected historical years.

Mapped area is polygon area, not a separate land-only measure. A small missing area need not have a small population. The instrument must still predict modern density and satisfy the assumption that historical settlement affects modern productivity only through the modeled channel after controls; coverage alone establishes neither.